

## No. of Functional Features Included in the Solution

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 29 June 2026                    |
| <b>Team ID</b>       | SWTID-2026-9188                 |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 5 Marks                         |

### Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

| S.No | Feature Name               | Feature Description                                                     | Module / Component     | Status | Marks Contribution |
|------|----------------------------|-------------------------------------------------------------------------|------------------------|--------|--------------------|
| 1    | AI Caption Generator       | Predicts whether a credit card application will be approved or rejected | Random Forest Model    | Done   | High               |
| 2    | Data Preprocessing         | Cleans and transforms applicant data before prediction                  | Data Processing Module | Done   | High               |
| 3    | Feature Encoding           | Converts categorical inputs into machine-readable format                | Encoding Module        | Done   | Medium             |
| 4    | User Input Form            | Allows users to enter applicant and financial details                   | Frontend UI            | Done   | Medium             |
| 5    | Input Validation           | Validates user inputs before processing                                 | Flask Backend          | Done   | Medium             |
| 6    | Risk Audit Report          | Displays prediction result with applicant risk summary                  | Report Generator       | Done   | Medium             |
| 7    | Prediction History Logging | Stores prediction details in JSON format for future reference           | Logging Module         | Done   | Low                |
| 8    | Responsive Web Interface   | Provides a mobile-friendly and responsive user interface                | HTML/CSS + Bootstrap   | Done   | Medium             |

Feature Summary:

| Metric                             | Count / Value |
|------------------------------------|---------------|
| Total Features Planned             | 8             |
| Total Features Implemented         | 8             |
| Core / Must-Have Features          | 5             |
| Additional / Nice-to-Have Features | 3             |
| Features Tested & Verified         | 8             |

Feature Category Breakdown:

| S.No | Category                  | Features in Category | Example Features                                          |
|------|---------------------------|----------------------|-----------------------------------------------------------|
| 1    | User Interface (UI)       | 3                    | User Input Form, Responsive UI, Prediction Result Display |
| 2    | Backend / Logic           | 3                    | Data Preprocessing, Input Validation, Prediction Engine   |
| 3    | Database / Storage        | 1                    | JSON Prediction History Logging                           |
| 4    | Machine Learning          | 1                    | Random Forest Credit Card Approval Prediction             |
| 5    | Security / Authentication | 1                    | Input Validation and Secure Request Handling              |